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Design and realization of wireless communication interface of portable single vehicle simulation tester

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Abstract

In order to solve the problems of complex circuit debugging and inconvenient operation, wireless communication technology is adopted and the USR-WIFI232-B2 module is used to complete the data transmission and reception between the single vehicle simulation tester and the upper computer. The hardware circuit and software flow of the portable single vehicle simulation tester are described, the wireless communication module is configured, and the laboratory test and on-site debugging are completed. It greatly facilitates the operation of the workers and improves the debugging efficiency.

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Keywords: single vehicle debugging; wireless communication; single vehicle simulation tester

1. Introduction

In the "Medium and Long-term Railway Network Planning", the China Railway Corporation takes the outline of a grand blueprint for the "eight horizontal and eight vertical" high-speed railway network in the new era as its primary goal¹. In order to meet the growing demand for the production of high-speed EMUs, major domestic locomotive factories must ensure both the output and the quality of the EMU products. Before the EMU leaves the factory, the debugging workers need to perform a series of complex inspections, but the debugging equipment is heavy and the wiring is complicated².

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At present, the domestic EMU debugging workshop adopts a movable workstation. This workstation is equipped with MVB network equipment and can be linked to the MVB network on the vehicle. It is indicated by the status of the simulated signal light in the display interface on the workbench. Whether the electrical circuit and the interface are connected correctly³. However, the existing debugging equipment needs to be connected to both ends of the vehicle to be tested, and the field wiring is messy, making it difficult for workers to operate⁴.

In order to solve the problems of complicated on-site wiring, messy wiring, and inconvenient operation by workers. According to actual needs, this paper designs a set of portable single vehicle simulation tester that can replace most of the lines on site through WiFi wireless network. The portable single vehicle simulation tester can greatly improve the efficiency of debugging, facilitate the operation of the workers and masters on the spot, and save the cost of various circuits on the spot.

2. The overall plan of single vehicle simulation tester design

In order to meet the complex debugging needs of the EMU, it solves the problems of messy wiring, complicated wiring and unchanging operation of existing single vehicle debugging equipment. The portable single vehicle simulation tester shall have a central control unit with multiple digital input and output channels; it shall have communication interfaces such as Ethernet, WiFi, CAN, and RS485, as shown in Figure 1.

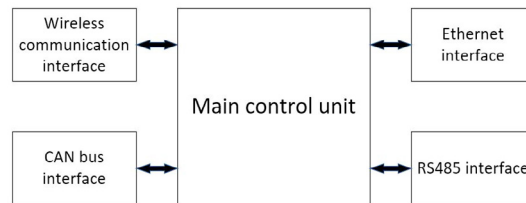


Fig. 1. The structure diagram of the communication interface of the single vehicle simulation tester

In the overall design scheme, the selection and design of related hardware are determined according to the functions of each part. Including the main control chip design, Ethernet module design, WiFi wireless communication module design, storage module design, power supply module design and so on. In this design, STM32F103ZET6 is the main control chip, and the USR-WIFI232-B2WiFi wireless communication module is connected through the serial port to complete the data communication between the on-site intelligent monitoring equipment and the single vehicle simulation tester. Expand CAN bus interface, etc. to meet different field conditions as data communication interface.

As shown in Figure 2. Four single vehicle simulation testers can communicate with monitoring equipment through WiFi wireless communication network, including industrial computer, industrial tablet computer and other mobile terminals to complete the data transmission and reception. When the wireless network is not good, you can communicate via CAN bus, connect the single vehicle simulation tester of other unit networks in a wired manner, and use the network cable to connect to the upper computer through the Ethernet interface to monitor the single vehicle in real time⁵.

This article mainly completes the wireless communication interface design. According to the requirement of single vehicle debugging and the occupation of pin resources by hardware expansion, the main control chip selects STM32F103ZET6. The main control chip FSMC and ADC converter channel resources are rich, the operating frequency can reach 72MHz, and the memory includes 512K bytes of flash memory and 64K bytes of SRAM high-speed memory. It can meet the requirements of the single vehicle simulation tester for interface resources. Its flexible design can better configure the pin resources of the main control chip, and can better expand the wireless communication module.

The wireless communication module uses the USR-WIFI232-B2 module, which transmits data through interfaces such as the UART serial port of the main control chip, and connects the user's physical device to the WiFi wireless

network⁶. The method of connecting the main control chip through the UART serial port can easily connect the single vehicle simulation tester to the WiFi wireless network, so that the intelligent monitoring equipment can control and collect the electrical switch values in the single vehicle simulation tester through the WiFi wireless network. The USR-WIFI232-B2 module has a maximum transmission distance of 280m, which can fully adapt to the working conditions of the site and effectively solve the problems of complex on-site wiring and messy wiring⁷.

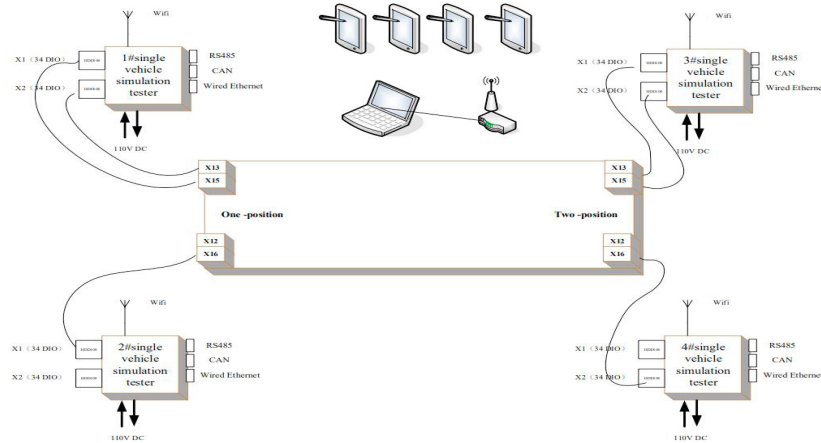


Fig. 2. Effect diagram of single vehicle debugging

3. The hardware design of the wireless communication interface of the single vehicle simulation tester

The four single vehicle simulation testers need to communicate with intelligent monitoring equipment through WiFi wireless communication network. As shown in the figure, the design of the wireless communication interface is to connect the central control unit STM32F103ZET6 and the wireless communication module USR-WIFI232-B2 through a UART serial port, and connect the single vehicle simulation tester to the WiFi wireless network to realize data communication. The wireless communication module handles the reception of control messages and the sending of status messages.

As shown in Figure 3, Circuit diagram of wireless communication module. USR-WIFI232-B2 needs 3.3V power supply, connected to READY, LINK two LED lights show that the wireless communication module is working normally when the system is running. And connect the reset and restore factory settings pins to the MCU to ensure that the wireless communication function of the system can be restored by resetting when the communication times out.

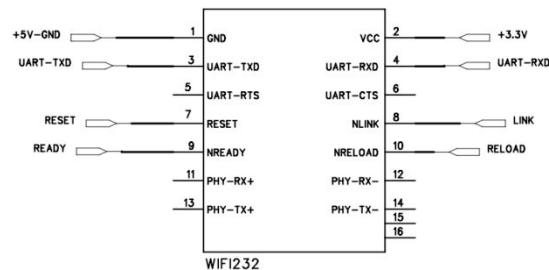


Fig. 3. Circuit diagram of wireless communication module.

4. Software design of wireless communication interface for single vehicle simulation tester

On the basis of the hardware design of the network communication interface of the single vehicle simulation tester, the software design of the network interface is mainly completed. The single vehicle simulation tester sends the electrical signals collected on the vehicle to be tested to the intelligent monitoring equipment, and the control signal of the intelligent monitoring equipment is transmitted to the single vehicle Simulation tester.

The network interface software system lies in the cooperation and coordination of the hardware to jointly complete the reception and transmission of control messages and status messages. The network interface program is a set of subroutines in the design of the single vehicle simulation tester. It is used for data transmission between the single vehicle simulation tester and the intelligent monitoring equipment. At the same time, it also has functions such as resetting the wireless communication module during communication timeout.

Figure 4 shows the wireless communication software flow. In the process of wireless communication data sending and receiving, the main control chip sends data to the wireless communication module through the UART serial port, waiting for data processing. Before sending to the intelligent monitoring device, prepare for sending and check the UART serial port data. Then you need to judge whether the frequency division is normal. If the communication timeout occurs, you can complete the reset of the wireless communication module through instructions, or restore the factory settings. To ensure that problems such as communication timeouts can be solved. After judging that the frequency division is normal, set the Ethernet port. According to the Ethernet application layer protocol, the wireless communication interface is set to port 503, and data is sent regularly. And pack the collected electrical switch into a status frame in the set Ethernet application layer protocol, and continue to send status messages. After each status message is sent, the control message of the intelligent monitoring device is received. As shown in the figure, this completes the sending and receiving of status messages and control messages. During the working process of the single vehicle simulation tester, the wireless communication module will continuously send status messages and receive control messages to ensure that the vehicle under test can obtain real-time and accurate data such as electrical switches.

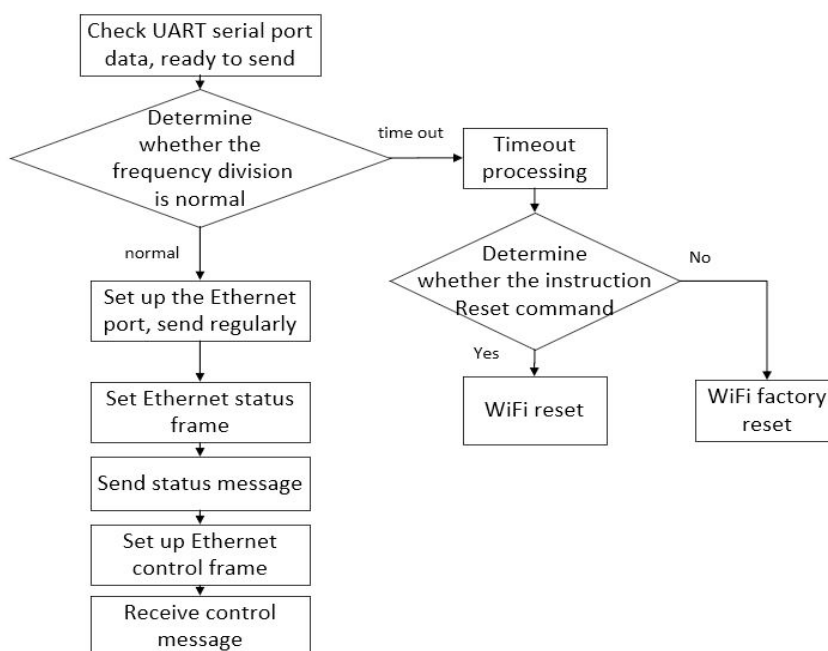


Fig. 4. Wireless communication software flow chart

The following is a part of the program for sending status messages written in C language. This program is to set the format of the status message, in which the two bytes "chRP1" are grabbed, and "chRP2" is the frame header of the status message; the high four bits of the third byte are inverted as the life signal, and the low four bits are used as the life signal. Own ID; after that is the status data with a length of 80 bytes; "chED1" and "chED2" are the end of the frame; finally Sum_Check is an accumulation and check function that does not contain itself to ensure the accuracy of the sent status message.

```

unsigned short index = 0,j,k;
S_Data_OUT_Buffer[index++] = chRP1;
S_Data_OUT_Buffer[index++] = chRP2;
liftSignal ^=0xf0;
S_Data_OUT_Buffer[index++] = liftSignal|Dev_ID;
S_Data_OUT_Buffer[index++] = 80;
S_Data_OUT_Buffer[index++] = chED1;
S_Data_OUT_Buffer[index++] = chED2;
S_Data_OUT_Buffer[index] = Sum_Check(S_Data_OUT_Buffer,index)

```

5. Laboratory testing and on-site debugging

In order to verify whether the portable single vehicle simulation tester is practical or not, first perform tests in the laboratory to verify the functions of the single vehicle simulation tester. Access the wireless communication module through a laptop computer to modify the parameters of the wireless communication module of the single vehicle simulation tester. Enter the configuration interface of the wireless communication module through the browser, change the AP mode to Station mode, and select the AP that the end tester needs to access in the pop-up AP list. In the wireless terminal settings, set the IP address to static (fixed) IP, and enter the IP address, subnet mask, and gateway setting information. Set static IP address, etc. as shown in Figure 5.

静态模式	
IP 地址	192.168.1.51
子网掩码	255.255.255.0
网关设置	192.168.1.1
域名服务器	

确定 取消

Fig. 5. Static IP address setting

In the serial port and network settings, set the baud rate of the serial port to 9600, and in the network parameter settings, set the network mode to server, the protocol to TCP, and the port to 503, and finally restart the module. In this way, the single vehicle simulation tester can automatically connect to the set WiFi network when it is turned on.

Set the network parameters as shown in Figure 6.



Fig. 6. Network parameter setting

Connect the single vehicle simulation tester with a cable to a circuit board that can observe the switch values through the LED lights, connect the wires to simulate the vehicle to be tested on site, and light up these electrical switch values one by one to ensure these switches. The amount of data is sent accurately and accurately according to the communication protocol



a



b

Fig. 7. (a) On-site debugging of the vehicle to be tested; (b) Upper computer debugging interface

Single vehicle debugging is to place two single vehicle simulation testers on the left and right ends of the vehicle to be tested, as shown in the figure. Each single vehicle simulation tester is connected to the vehicle to be tested through a cable. When collecting status data, the four single vehicle simulation testers collect the status data on the vehicle to be tested through the connected cable, and the four single vehicle simulation testers are connected to the WiFi wireless communication network and sent to the upper computer (that is the intelligent monitoring device). State data collected by the single vehicle simulation tester.

6. Conclusion

The portable single vehicle simulation tester is based on STM32F103ZET6 as the central control unit, adopts modular design, expands wireless communication interface, Ethernet interface, CAN bus interface and so on. The different modules are interconnected by backplane serial bus, which can adapt to different types of vehicle debugging. Compared with the existing EMU debugging equipment, this design is more suitable for long-distance

train debugging. Its intelligent monitoring equipment can remotely control the output of the end unit through the WiFi wireless communication network and collect the input signal of the end unit. Because the design adopts WiFi wireless communication network, it can reduce the lines of on-site connection without manual operation, reduce the walking distance of debugging, and greatly improve the efficiency of debugging.

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